

Synchrotron Light Source Stabilization via Model-Independent Machine Learning

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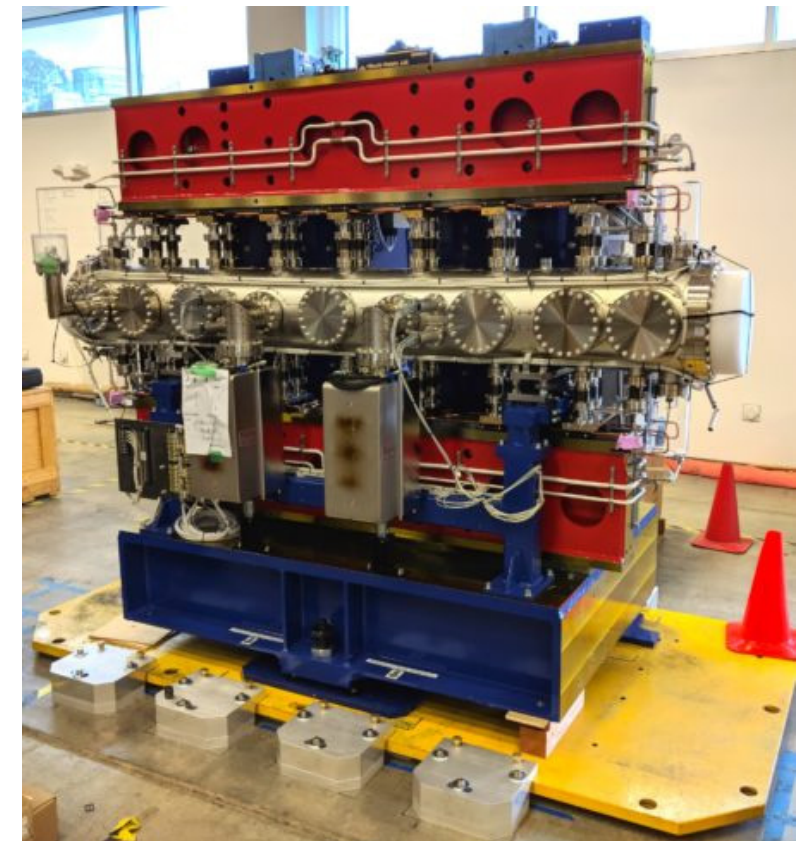


The Need for Stabilization

- At the ALS and various other High-Brightness Light Sources, source position/angle and intensity stability must be provided so users are able to collect high quality data from various x-ray experiments.
- Future Ultra-High Light Source upgrades will provide orders of magnitude higher brightness and transverse coherence, enabling the most demanding experiments.

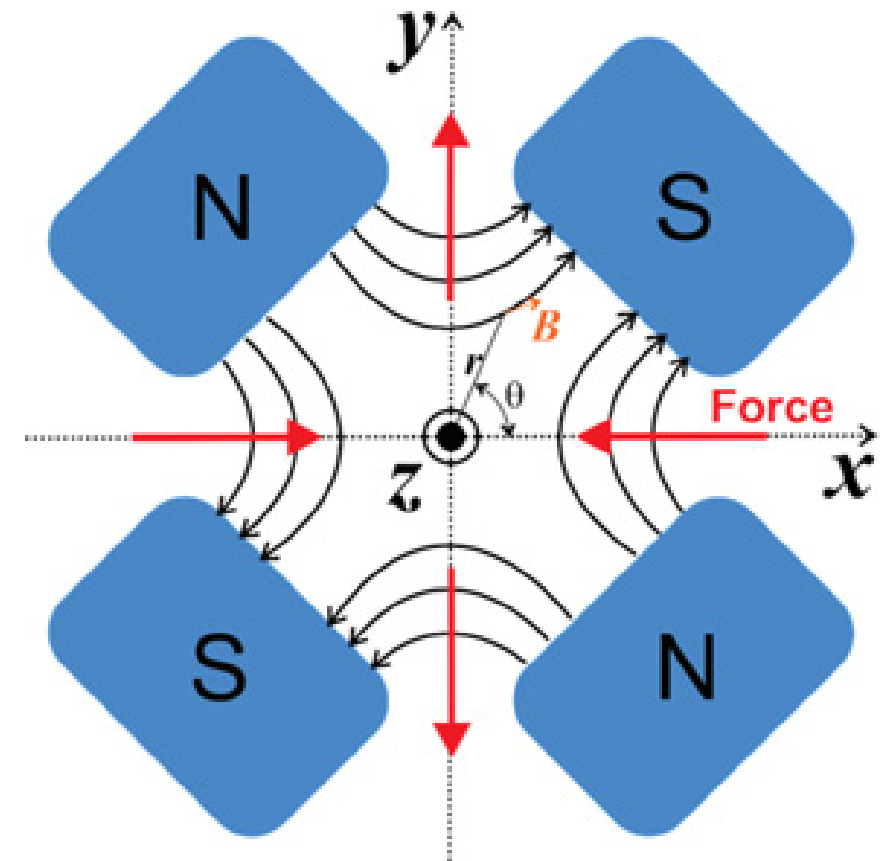
Causes of Beam Size Fluctuations

- ID gap/phase changes by users during operation.
- Misalignment of skew quadrupoles
 - Vibrations
- Drift
 - Thermal, ground motion
- Current loss due to standard operations
 - Countered by top-off injections

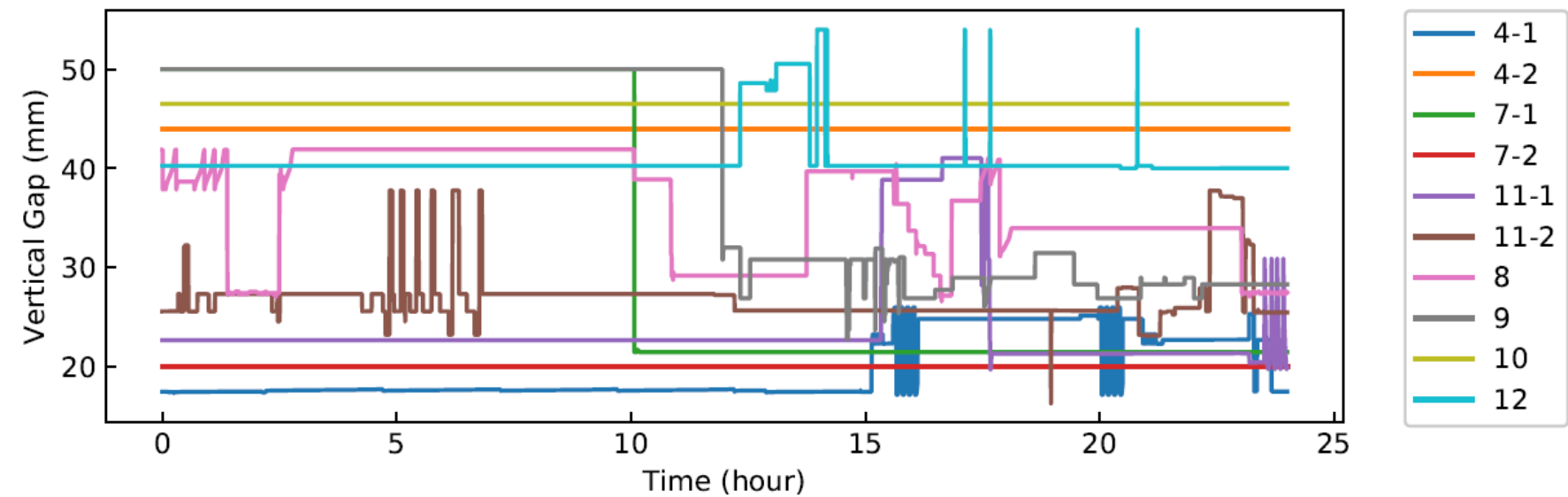
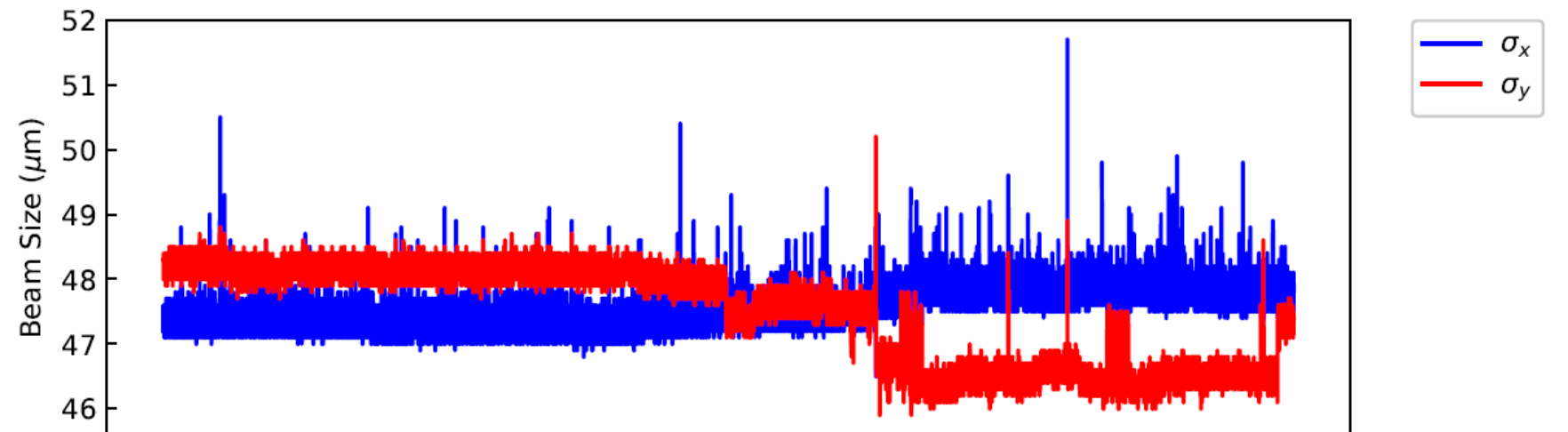


Current Correction Methods

- Orbit Feedbacks
- Local skew quadrupole corrections
- ID FF tables correct systematic focusing and skew quadrupole errors.
 - Very time consuming to collect data and develop tables
- Over 24-hour user shift, vertical beam size changes even with corrections in place. Due to ID gap/phase changes.

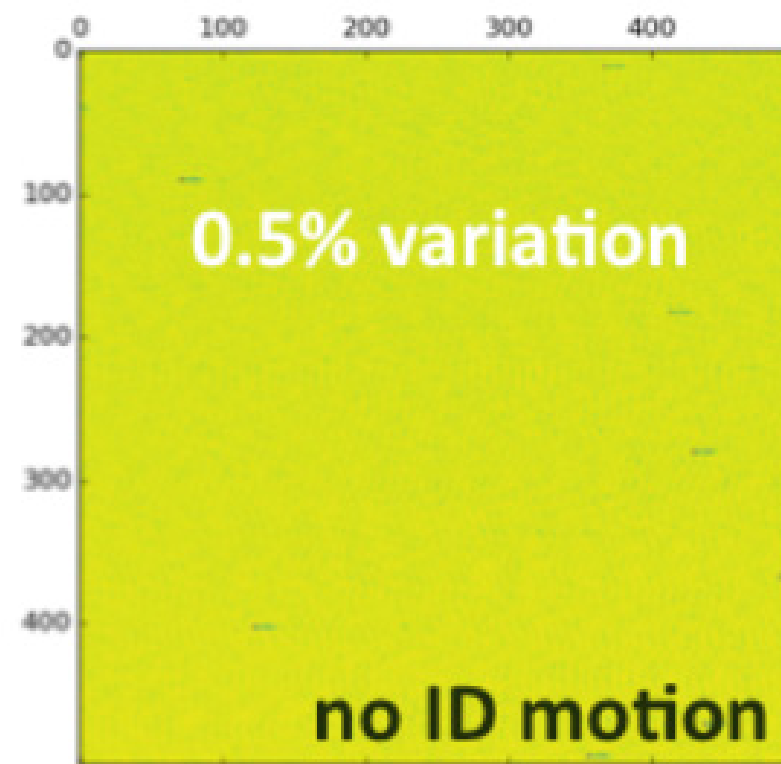
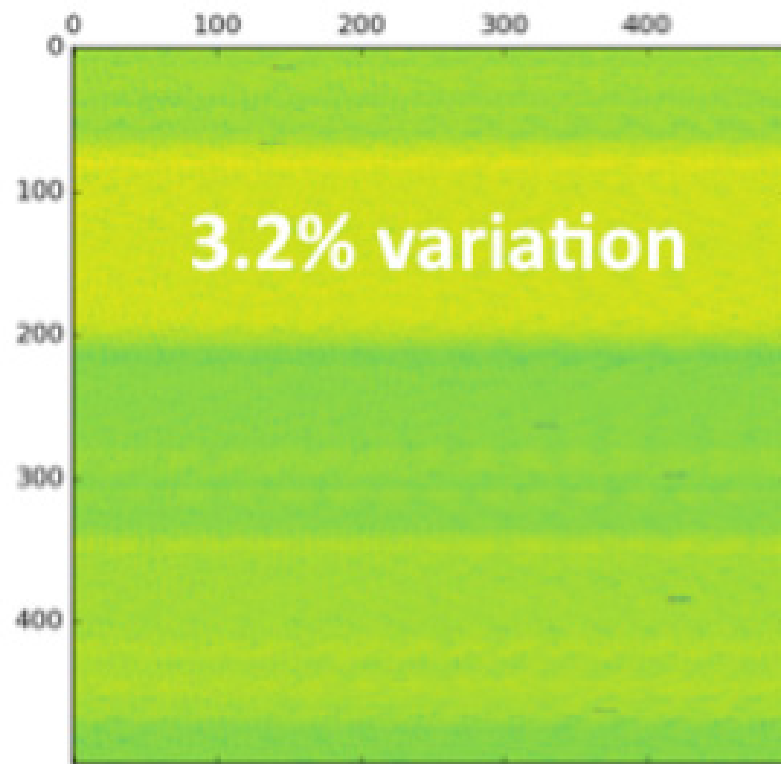


ALS Beam Data



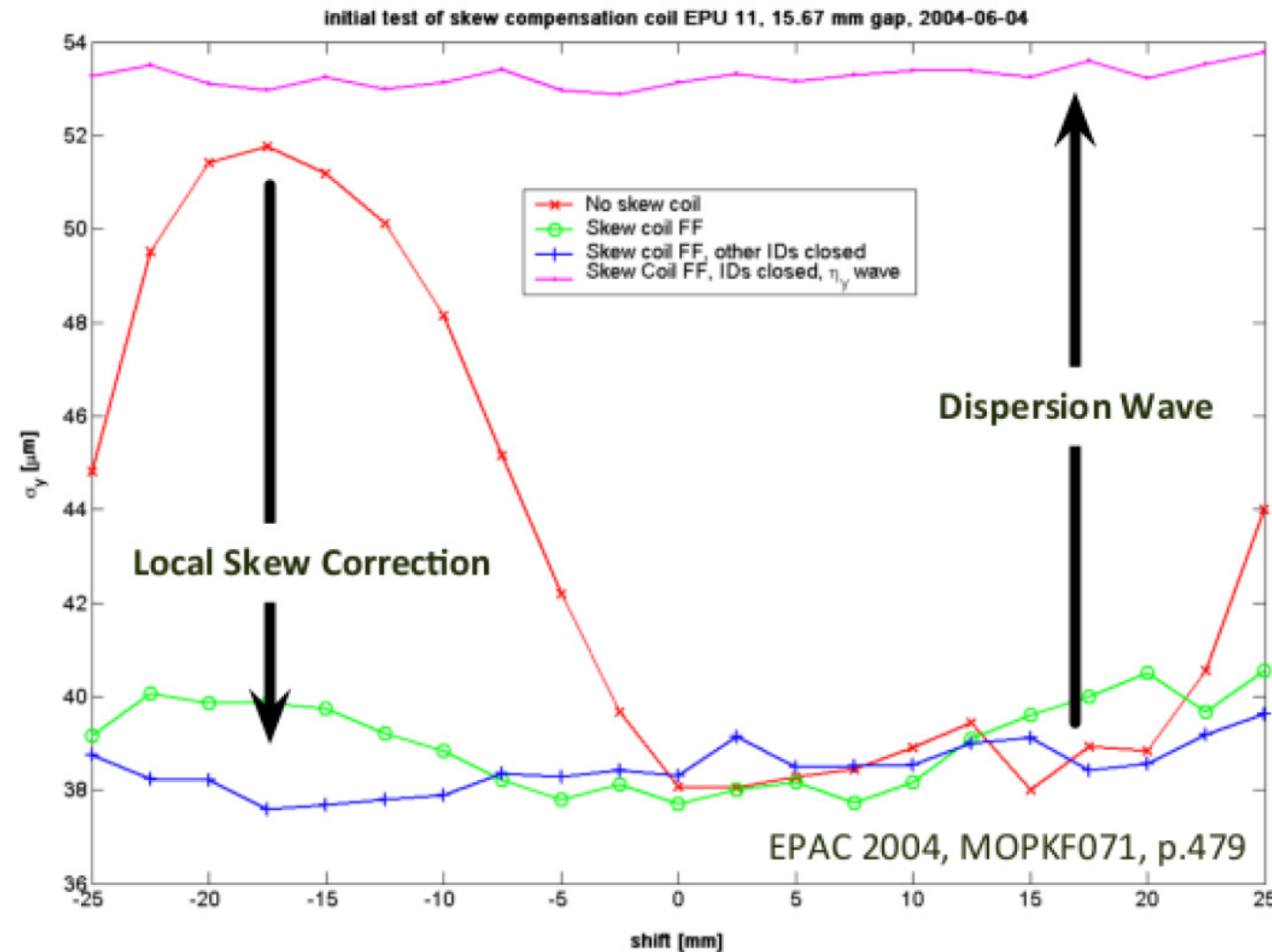
Direct Affect on User Data

ALS
beamline
5.3.2.2
(STXM)



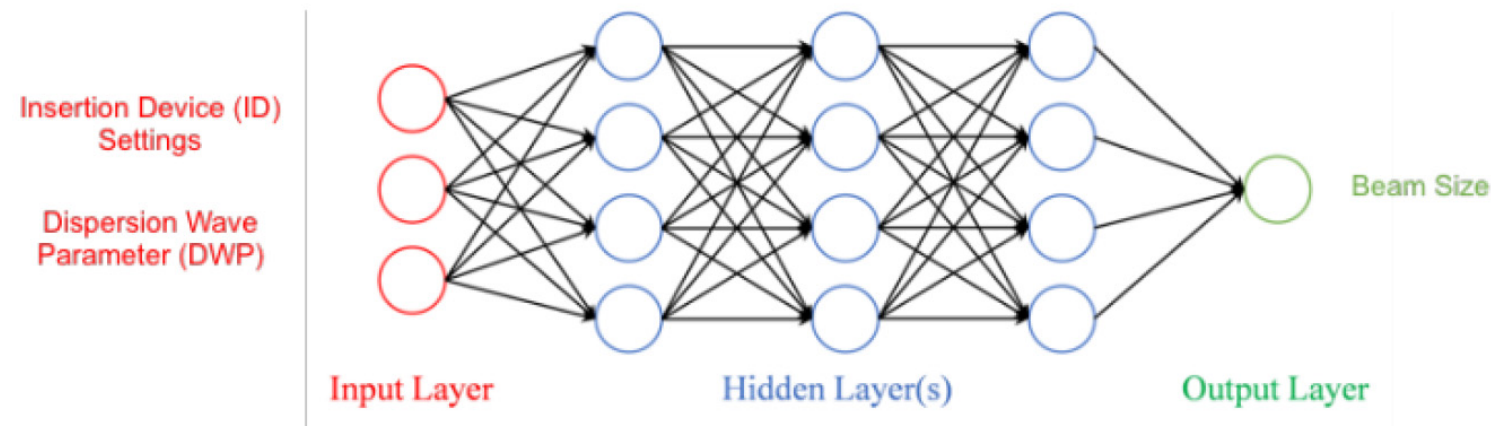
Dispersion Wave Parameter

Controls level of vertical dispersion due to effect of skew quadrupoles



Neural Network Training

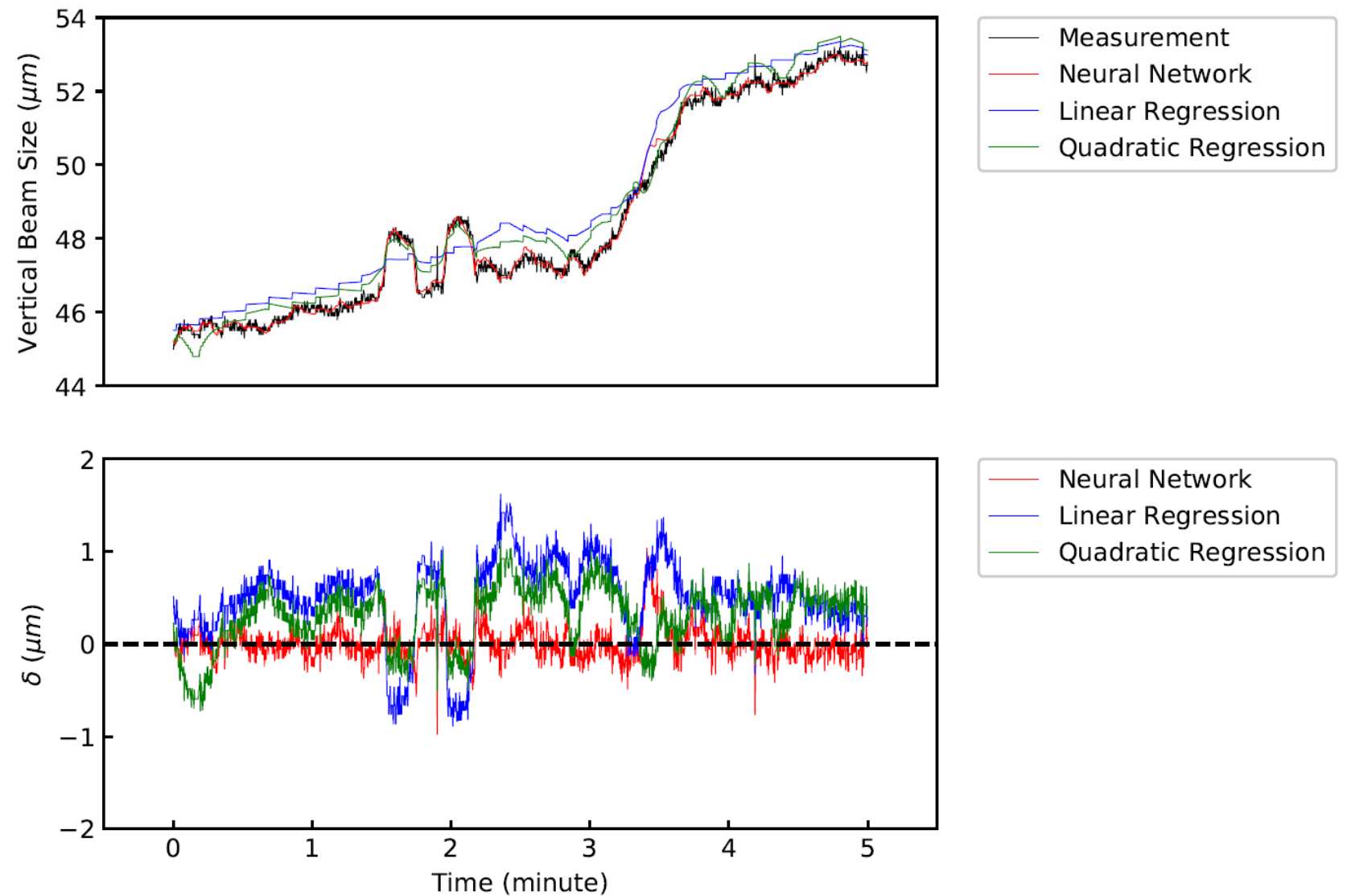
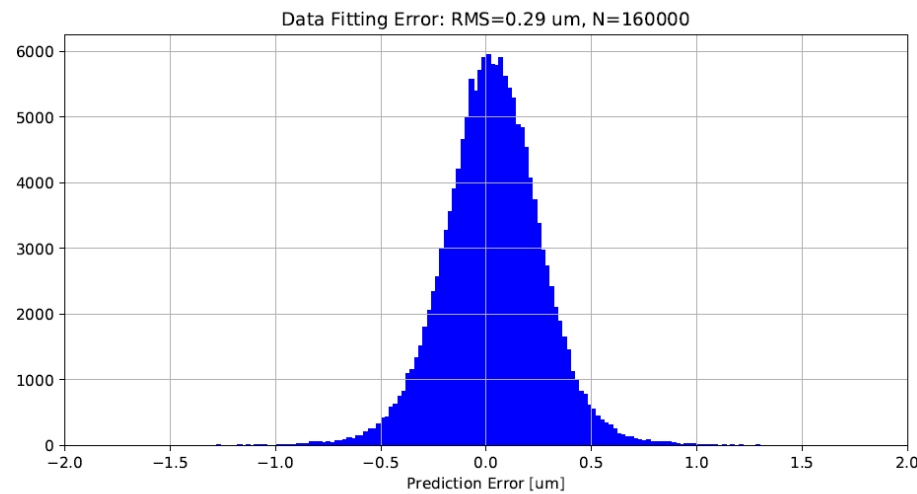
- NN can deliver accurate predictions for beam size as a function of ID gap/phase settings and DWP.
- NN is trained using 10 Hz data including all ID and skew configurations.
 - Also includes measured beam size
- Training time: ~ 10 minutes



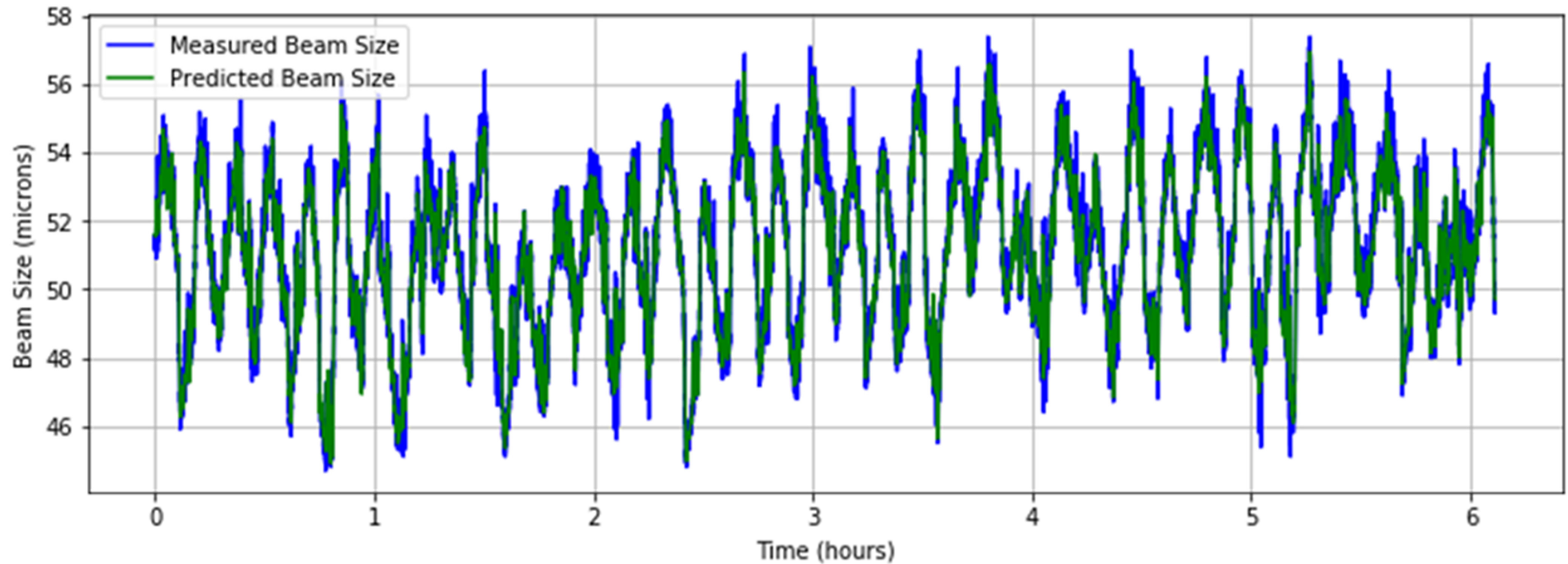
- 3 layers
 - 128,64,32
- Dropout = 0.2
- L2 Regularizer = 10^{-4}

Neural Network Validation

- Compared to both linear and quadratic regressions.

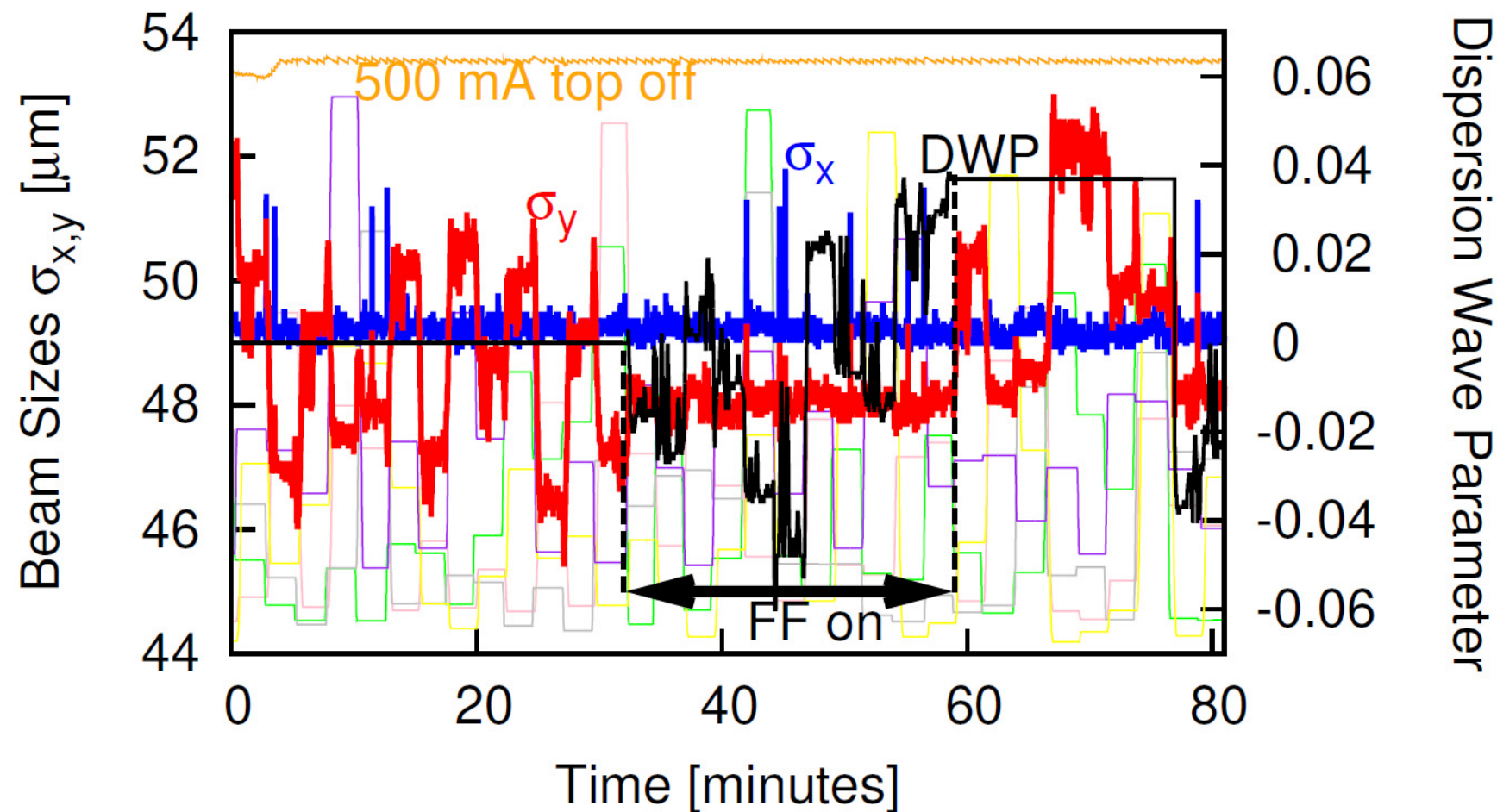


Neural Network Training Results



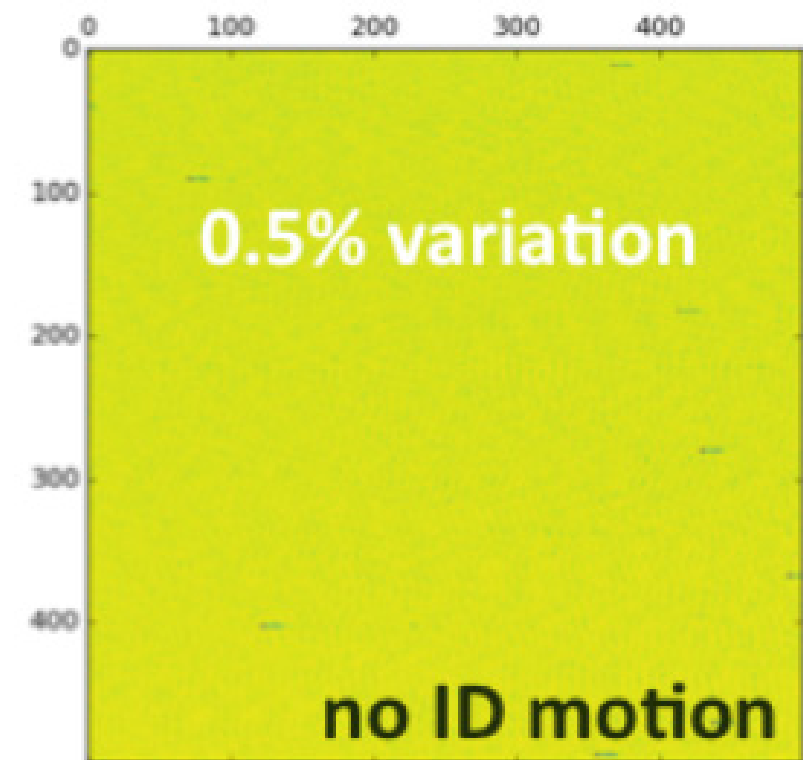
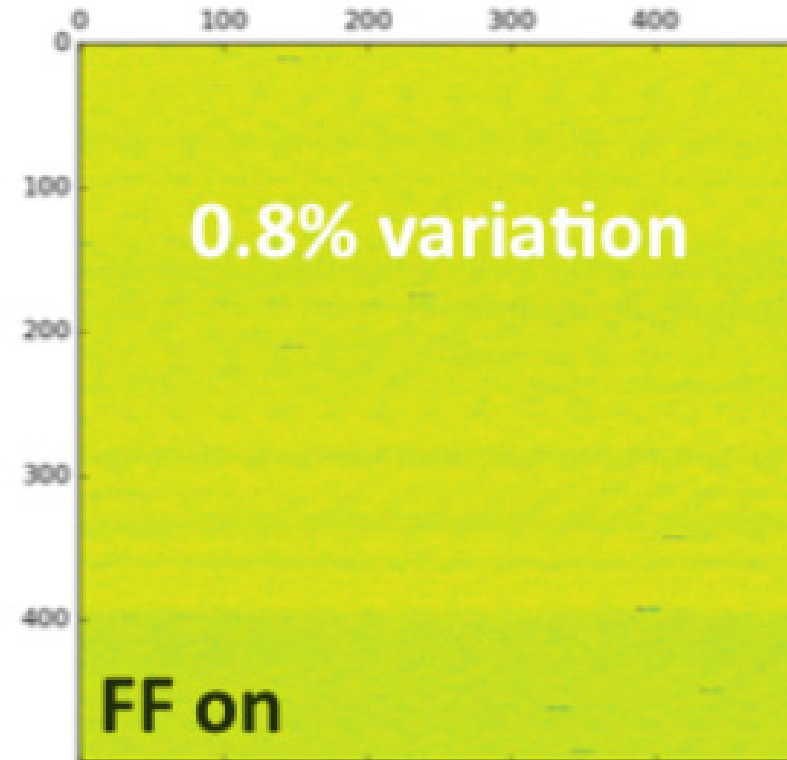
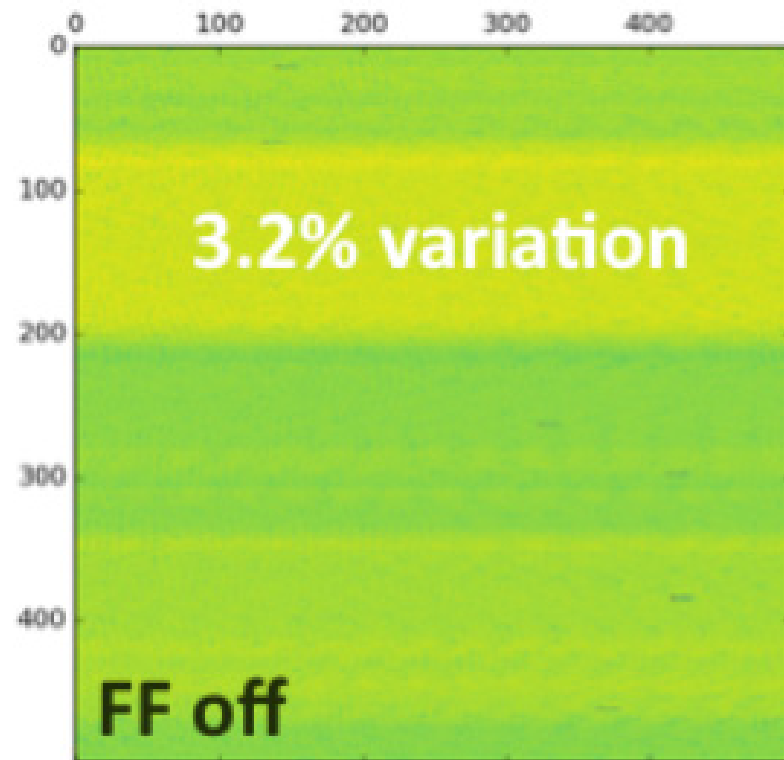
NN used during User-Ops

- We scan all user-configured ID settings in real time.
- Numerically sweep through entire DWP range.
- Choose DWP that produces beam size closest to target.



NN affect on STXM data

- Due to vertical beam size stabilization, STXM noise level decreases by a factor of 4.



Conclusion

- NN has been shown to successfully learn non-linear relations between ID gap/phase settings, DPW, and beam size.
- NN-based FF systems have been demonstrated dramatically stabilize beam size during user-ops
 - Results in dramatic decrease in noise level of data collected.
- RMS stability was improved over extended periods of time without further dedicated machine time or model retraining.
- Future work will explore NN hyperparameter space to further tune properties of the NN.
- Focus on varying online-retraining experiments

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